MODULE *FileTransfer*

Model of send-file-over-network $V0$: src/*leader.py* and src/*follower.py*.
send(2) failures are modeled as message loss, *recv*(2) failures as timeouts.

VARIABLES

lState,	Leader state, *LeaderState* in *leader.py*
fState,	Follower state, *FollowerState* in *follower.py*
restartHandled,	Follower's volatile *restart_handled* flag
fResult,	result the Follower will put in its *BlobReply*
fileOnDisk,	TRUE once the Follower's disk durably holds the file
msgs	messages in the network, in both directions

$vars \triangleq \langle lState, fState, restartHandled, fResult, fileOnDisk, msgs \rangle$

BlobReply results: 0, *ENOMEM* (retryable) and any other errno (terminal).

$Results \triangleq \{\text{"ok"}, \text{"retry"}, \text{"fatal"}\}$

$Messages \triangleq$
$\{[type \mapsto \text{"RestartedQuery"}], [type \mapsto \text{"Blob"}]\}$
$\cup [type : \{\text{"RestartedReply"}\}, restarted : \text{BOOLEAN}\ ]$
$\cup [type : \{\text{"BlobReply"}\}, result : Results]$

$Init \triangleq$
$\land\ lState = \text{"RESTARTED_LOOP"}$
$\land\ fState = \text{"WAITING_FOR_MESSAGE"}$
$\land\ restartHandled = \text{FALSE}$
$\land\ fResult = \text{"none"}$
$\land\ fileOnDisk = \text{FALSE}$
$\land\ msgs = \{\}$

Leader

$LeaderSendRestartedQuery \triangleq$
$\land\ lState = \text{"RESTARTED_LOOP"}$
$\land\ lState' = \text{"RESTARTED_SENT"}$
$\land\ msgs' = msgs \cup \{[type \mapsto \text{"RestartedQuery"}]\}$
$\land\ \text{UNCHANGED}\ \langle fState, restartHandled, fResult, fileOnDisk \rangle$

$LeaderRecvRestartedReply \triangleq$
$\land\ lState = \text{"RESTARTED_SENT"}$
$\land\ \exists\, m \in msgs :$
$\quad \land\ m.type = \text{"RestartedReply"}$
$\quad \land\ lState' = \text{IF}\ m.restarted\ \text{THEN}\ \text{"BLOB_LOOP"}\ \text{ELSE}\ \text{"FINAL"}$
$\land\ \text{UNCHANGED}\ \langle fState, restartHandled, fResult, fileOnDisk, msgs \rangle$

$LeaderSendBlob \triangleq$
$\land\ lState = \text{“BLOB_LOOP”}$
$\land\ lState' = \text{“BLOB_SENT”}$
$\land\ msgs' = msgs \cup \{[type \mapsto \text{“Blob”}]\}$
$\land$ UNCHANGED $\langle fState, restartHandled, fResult, fileOnDisk \rangle$

$LeaderRecvBlobReply \triangleq$
$\land\ lState = \text{“BLOB_SENT”}$
$\land\ \exists\, m \in msgs :$
$\quad\land\ m.type = \text{“BlobReply”}$
$\quad\land\ lState' = \text{CASE}\ m.result = \text{“ok”} \rightarrow \text{“FINAL”}$
$\qquad\qquad\qquad\square\ \ m.result = \text{“retry”} \rightarrow \text{“BLOB_LOOP”}$
$\qquad\qquad\qquad\square\ \ m.result = \text{“fatal”} \rightarrow \text{“FAILED”}$
$\land$ UNCHANGED $\langle fState, restartHandled, fResult, fileOnDisk, msgs \rangle$

Also covers $recv(2)$ failing while waiting for a reply.
$LeaderTimeout \triangleq$
$\land\ lState \in \{\text{“RESTARTED_SENT”}, \text{“BLOB_SENT”}\}$
$\land\ lState' = \text{IF}\ lState = \text{“RESTARTED_SENT”}\ \text{THEN “RESTARTED_LOOP” ELSE “BLOB_LOOP”}$
$\land$ UNCHANGED $\langle fState, restartHandled, fResult, fileOnDisk, msgs \rangle$

$LeaderCrash \triangleq$
$\land\ lState \notin \{\text{“FINAL”}, \text{“FAILED”}\}$
$\land\ lState' = \text{“RESTARTED_LOOP”}$
$\land$ UNCHANGED $\langle fState, restartHandled, fResult, fileOnDisk, msgs \rangle$

Follower

$FollowerRecvRestartedQuery \triangleq$
$\land\ fState = \text{“WAITING_FOR_MESSAGE”}$
$\land\ [type \mapsto \text{“RestartedQuery”}] \in msgs$
$\land\ fState' = \text{“RESTARTED_REPLY_LOOP”}$
$\land$ UNCHANGED $\langle lState, restartHandled, fResult, fileOnDisk, msgs \rangle$

$FollowerSendRestartedReply \triangleq$
$\land\ fState = \text{“RESTARTED_REPLY_LOOP”}$
$\land\ fState' = \text{“WAITING_FOR_MESSAGE”}$
$\land\ msgs' = msgs \cup \{[type \mapsto \text{“RestartedReply”}, restarted \mapsto \neg restartHandled]\}$
$\land$ UNCHANGED $\langle lState, restartHandled, fResult, fileOnDisk \rangle$

$FollowerRecvBlob \triangleq$
$\land\ fState = \text{“WAITING_FOR_MESSAGE”}$
$\land\ [type \mapsto \text{“Blob”}] \in msgs$
$\land\ fState' = \text{“WRITING”}$
$\land$ UNCHANGED $\langle lState, restartHandled, fResult, fileOnDisk, msgs \rangle$

$FollowerWriteCompleted \triangleq$
$\land\ fState = \text{“WRITING”}$
$\land\ fState' = \text{“BLOB_REPLY_LOOP”}$
$\land\ fileOnDisk' = \text{TRUE}$
$\land\ restartHandled' = \text{TRUE}$
$\land\ fResult' = \text{“ok”}$
$\land\ \text{UNCHANGED}\ \langle lState,\ msgs \rangle$

The write is atomic (write, fsync, rename), so a failure leaves the disk as it was.

$FollowerWriteFailed \triangleq$
$\land\ fState = \text{“WRITING”}$
$\land\ fState' = \text{“BLOB_REPLY_LOOP”}$
$\land\ fResult' \in \{\text{“retry”},\ \text{“fatal”}\}$
$\land\ \text{UNCHANGED}\ \langle lState,\ restartHandled,\ fileOnDisk,\ msgs \rangle$

$FollowerSendBlobReply \triangleq$
$\land\ fState = \text{“BLOB_REPLY_LOOP”}$
$\land\ fState' = \text{“WAITING_FOR_MESSAGE”}$
$\land\ msgs' = msgs \cup \{[type \mapsto \text{“BlobReply”},\ result \mapsto fResult]\}$
$\land\ fResult' = \text{“none”}$
$\land\ \text{UNCHANGED}\ \langle lState,\ restartHandled,\ fileOnDisk \rangle$

Volatile state is lost; the file on disk survives.

$FollowerCrash \triangleq$
$\land\ fState' = \text{“WAITING_FOR_MESSAGE”}$
$\land\ restartHandled' = \text{FALSE}$
$\land\ fResult' = \text{“none”}$
$\land\ \text{UNCHANGED}\ \langle lState,\ fileOnDisk,\ msgs \rangle$

Network: receiving does not remove a message, which models duplication.

$LoseMessage \triangleq$
$\land\ \exists\, m \in msgs : msgs' = msgs \setminus \{m\}$
$\land\ \text{UNCHANGED}\ \langle lState,\ fState,\ restartHandled,\ fResult,\ fileOnDisk \rangle$

$Next \triangleq$
$\lor\ LeaderSendRestartedQuery$
$\lor\ LeaderRecvRestartedReply$
$\lor\ LeaderSendBlob$
$\lor\ LeaderRecvBlobReply$
$\lor\ LeaderTimeout$
$\lor\ LeaderCrash$
$\lor\ FollowerRecvRestartedQuery$
$\lor\ FollowerSendRestartedReply$

$\lor$ *FollowerRecvBlob*
$\lor$ *FollowerWriteCompleted*
$\lor$ *FollowerWriteFailed*
$\lor$ *FollowerSendBlobReply*
$\lor$ *FollowerCrash*
$\lor$ *LoseMessage*

$Spec \triangleq Init \land \Box[Next]_{vars}$

Properties

$TypeOK \triangleq$
$\land$ $lState \in \{$"RESTARTED_LOOP", "RESTARTED_SENT", "BLOB_LOOP", "BLOB_SENT", "FINAL", "FAILED"$\}$
$\land$ $fState \in \{$"WAITING_FOR_MESSAGE", "RESTARTED_REPLY_LOOP", "WRITING", "BLOB_REPLY_LOOP"$\}$
$\land$ $restartHandled \in$ BOOLEAN
$\land$ $fResult \in Results \cup \{$"none"$\}$
$\land$ $fileOnDisk \in$ BOOLEAN
$\land$ $msgs \subseteq Messages$

The Leader only reports success if the Follower really holds the file.
$FinalImpliesFile \triangleq lState =$ "FINAL" $\Rightarrow fileOnDisk$

The Follower never claims to hold the file when it does not.
$RepliesAreTruthful \triangleq$
$\land [type \mapsto$ "RestartedReply", $restarted \mapsto$ FALSE$] \in msgs \Rightarrow fileOnDisk$
$\land [type \mapsto$ "BlobReply", $result \mapsto$ "ok"$] \in msgs \Rightarrow fileOnDisk$

Not in the *.cfg*: *TLC* finds a counterexample (see the trace it reports).
$FailureIsGenuine \triangleq lState =$ "FAILED" $\Rightarrow \neg fileOnDisk$
